The Karhunen-Loeve Expansion and Mercers Theorem

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Let's recall the KL expansion from [1, 2.3.3].4

1 Karhunen-Loeve Expansion

Differentiating equation (2.26) with respect to $h(x)$ and setting the result equal to zero, gives

$$\frac{\mathrm{d}F}{\mathrm{d}\,h(x)} = \int_D \int_D R_{ww}(x_1, x_2)\, h(x_1)\, d\,x_1 - \lambda\, h(x_2)\, d\,x_2 = 0 \tag{1}$$

which is satisfied when

$$\int_D R_{ww}(x_1, x_2)\, h(x_2)\, d\,x_2 = \lambda\, h(x_1) \tag{2}$$

1.1 Uniqueness of the Expansion

The random variables appearing in an expansion of the kind given by equation (2.10) and the orthogonality and orthonormal functions $\{f_n(x)\}$ and the constants $\{\lambda_n\}$ are respectively the eigenfunctions and the eigenvalues of the covariance kernel as given by equation (2.8).

The "if" part is an immediate consequence of equation (2.11). To show the "only if" part, equation (2.12) can be used with $\langle \xi_n, \xi_m \rangle = \delta_{nm}$ to obtain

$$C(x_1, x_2) = \sum_{n=0}^{\infty} \lambda_n\, f_n(x_1)\, f_n(x_2) \tag{3}$$

In addition to the mean-square error minimizing property the Karhunen-Loeve expansion has the minimum representation entropy out of all possible representations.

1.2 Solving Integral Covariance Operator Eigenfunction Equations With The KL Expansion

The Karhunen-Loeve expansion is an instrumental part of solutions of the integral covariance operator eigenfunction equation

$$\int_D C(s, t)\, f_n(t)\, d\,t = \lambda_n\, f_n(s) \tag{4}$$

where $C(s,t)$ is an autocovariance function and n is a non-negative integer. Equation (2.33) is a homogeneous Fredholm integral equation of the second kind. . Being an autocovariance function, the kernel $C(x_1, x_2)$ is bounded, symmetric, and positive definite. This fact simplifies the ensuing analysis considerably by guaranteeing a number of properties for the eigenfunctions and the eigenvalues that are solutions to equation (2.33). Specifically:

1. The set $\{f_n(x)\}$ of eigenfunctions is orthogonal and complete.
2. For each eigenvalue λ_n, there correspond at most a finite number of linearly independent eigenfunctions.
3. There are at most a countably infinite set of eigenvalues.
4. The eigenvalues are all positive real numbers.
5. The kernel $C(s,t)$ admits the following uniformly convergent expansion:

$$C(x_1, x_2) = \sum_{k=1}^{\infty} \lambda_k f_k(s) f_k(t) \tag{5}$$

The Karhunen-Loeve expansion of a process was derived based on the preceding analytical properties of its covariance function. These properties are independent of the stochastic nature of the process involved, which allows the expansion to be applied to a wide range of processes including nonstationary and multidimensional processes. [1, 2.3.3]

2 Mercers Theorem

Let $\{X(t), t \in \mathbb{R}\}$ be a second order (scalar) process with mean function

$$m(t) = \mathbb{E}[X(t)] \tag{6}$$

and a continuous covariance function

$$(s,t) \mapsto r(s,t) = \mathbb{E}\left[(X(s) - m(s))\,(X(t) - m(t))\right] \tag{7}$$

Then $r(s,t)$ is positive definite, and if r is a (temporarily, assumed hereafter) continuous, by the classical Mercer theorem[3, p.245] since

$$\int_{\mathbb{R}} \int_{\mathbb{R}} |r(s,t)|^2 \, d\,s\,d\,t < \infty \tag{8}$$

it can be represented by a uniformly convergent series: ('bar' for complex conjugate)

$$r(s,t) = \sum_{i=1}^{\infty} \frac{\phi_i(s)\overline{\phi_i(t)}}{\lambda_i} \forall \lambda_i > 0 \tag{9}$$

where the $\phi_i(\cdot)$ are continuous functions satisfying the integral equation

$$\phi_i(t) = \lambda \int_{\mathbb{R}} r(s,t)\, \phi_i(s)\, d\, s \tag{10}$$

and $\sum_{i=1}^{\infty} \frac{1}{\lambda_i} < \infty$. Here the λ_i are the eigenvalues (counted according to their multiplicity) and ϕ_i are the corresponding eigenfunctions of the "kernel" r, and $\{\phi_n, n \geq 1\}$ forms a complete orthonormal set in the Lebesgue space $L^2(\mathbb{R})$, with Lebesgue measure, satisfying (15). This classical result and its relation to the Hilbert-Schmidt theory of symmetric kernels is nicely treated in the above reference, and their properties are needed here. First we consider the case that $\mathbb{E}[X(t)] = m(t) = 0$ so that the $X(t)$ are centered. Now define the random variables

$$\xi_n = \sqrt{\lambda_n} \int_{\mathbb{R}} X(t) \overline{\phi_n(t)} d\, t \forall n \geq 1 \tag{11}$$

where the integral is obtained using Fubini's theorem, since $X(t,\omega)$ is jointly measurable ($r(\cdot,\cdot)$ being jointly continuous) in (t,ω). [Alternatively it may be regarded as a Bochner integral.] In any case, we get

$$\begin{aligned} \mathbb{E}\left[\xi_n\, \xi_m\right] &= \sqrt{\lambda_n\, \lambda_m} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{E}\left[X(s)\, X(t)\right] \overline{\phi_n(s)} \phi_m(t)\, d\, s\, d\, t \\ &= \sqrt{\lambda_n\, \lambda_m} \iint_{\mathbb{R}^2} r(s,t) \overline{\phi_n(s)} \phi_m(t)\, d\, s\, d\, t \\ &= \frac{\delta_{nm}}{\sqrt{\lambda_m}} \int_{\mathbb{R}} \overline{\phi_n(t)} \phi_m(t)\, d\, t \end{aligned} \tag{12}$$

It follows (on expanding inner products) that $X_n(t) = \sum_{k=1}^{\infty} \xi_k\, \phi_k(t) / \sqrt{\lambda_k} \to X(t)$ in $L^2(P)$, by (14), where $X(t) = \sum_{n=1}^{\infty} \xi_n\, \phi_n(t) / \sqrt{\lambda_n}$, and conversely if $X(t)$ is given by this series, converging in mean, then $E\left(X(s)\, X(t)\right) = \lim_{n\to\infty} \sum_{i=1}^{n} \xi_i \bar{\xi}_i \phi_i(s)\, \phi_i(t) / \lambda_i$ holds. If $E(X(t)) = m(t) \neq 0$, then the above argument applied to $Y(t) = X(t) - m(t)$ establishes the following classical Karhunen-LoÃ¨ve representation:

Proposition. If $\{X(t), t \in I\}$ is a second order process with $E(X(t)) = m(t)$, and a continuous covariance function $r(\cdot,\cdot)$ on a compact interval I, then

$$X(t) = m(t) + \sum_{n=1}^{\infty} \xi_n \frac{\phi_n(t)}{\sqrt{\lambda_n}} \forall t \in I \tag{13}$$

holds uniformly in t, and the convergence is in $L^2(P)$ where the $\lambda_n > 0$ and ϕ_n are the eigenvalues and the corresponding (complete orthonormal in $L^2(I)$) eigenfunctions of the kernel, satisfying (15), and hence the ξ_n are orthonormal in $L^2(P)$, given by (16). [2, ...]

Bibliography

[1] Roger G. Ghanem and Pol D. Spanos. *Stochastic finite elements: a spectral approach*. Springer-Verlag, Berlin, Heidelberg, 1991.

[2] Malempati M. Rao. *Stochastic Processes: Inference Theory*. Springer Monographs in Mathematics. Springer, 2nd edition, 2014.

[3] Frigyes Riesz and Béla Sz.-Nagy. *Functional Analysis*. Dover Books on Mathematics. Dover Publications, 2012.